

Advanced Flying Robots – Final Results

INDI Across Three Platforms: Standard, Thrust-Upgraded, Brushless CF21BL

Georg Wolnik

Multi Robot Systems Project
Mentors: Omar Elsayed, Khaled Wahba
Supervisor: Prof. Wolfgang Hönig

Final Presentation

Agenda

- ➊ Recap: planned trajectories + geometric vs. INDI (standard drone)
- ➋ Core result: figure-8 across all four modes
- ➌ Upgraded vs. brushless: oval, circle + kt sweeps
- ➍ Inverted loop
- ➎ Videos / demonstration
- ➏ Summary

From the last progress presentation

- Minimum-snap polynomial trajectories; two timing modes: Mode 1 (k_t timing), Mode 3 (time redistribution).
- Differential flatness maps the planned trajectory to the controller feedforward (thrust, attitude, $\dot{\omega}_d$).
- INDI (Tal & Karaman 2020): incremental, sensor-based control law, benchmarked on standard CF2.1.

Motion Planning — The Idea

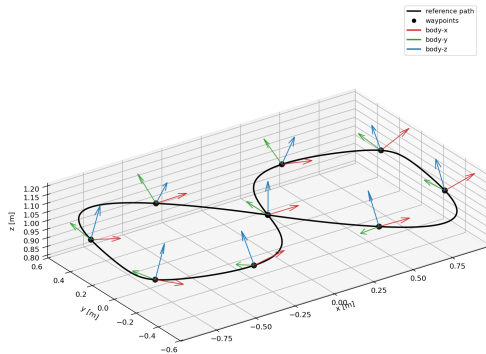
Core idea, in one sentence

Turn a handful of waypoints into **one smooth, closed-form path** — and get the drone's full attitude and thrust commands **for free** from that same path, instead of planning them separately.

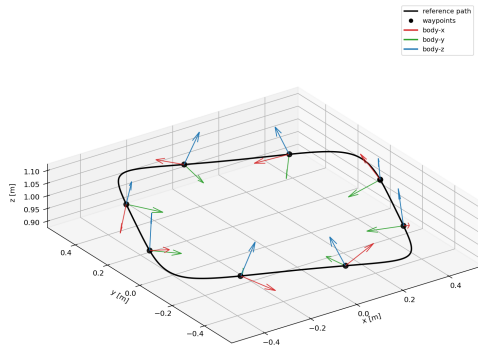
- **Problem:** waypoints alone are not a flyable trajectory — need something smooth, dynamically feasible, and exactly repeatable, not just “connect the dots.”
- **Minimum-snap polynomials** (Mellinger & Kumar 2011): fit a piecewise polynomial through the waypoints minimizing snap (4th derivative of position), continuous up to snap at every segment boundary — snap drives *required thrust and attitude rate*, so smoother snap means gentler, more achievable commands.
- **Differential flatness** (Mellinger & Kumar 2011; rotor-drag extension: Faessler et al. 2018): the quadrotor's full state — attitude, body rate, thrust — is recovered in *closed form* from the position path and its derivatives plus yaw. No attitude has to be authored by hand.
- **Full feedforward for every controller:** geometric *and* INDI both get thrust, attitude, body rate, and angular acceleration from the same path — enabling a fair, apples-to-apples comparison across controllers and platforms.

Planned Trajectories — Figure-8 / Circle / Corner

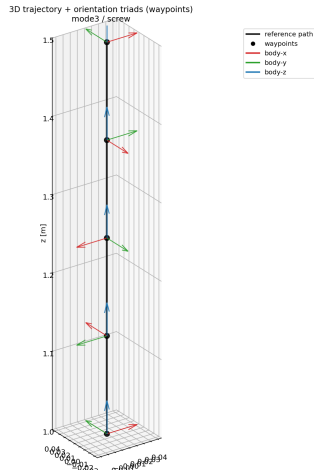
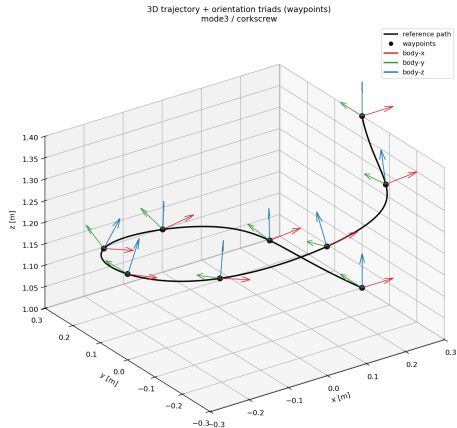
3D trajectory + orientation triads (waypoints)
model / figure8



3D trajectory + orientation triads (waypoints)
mode3 / circle

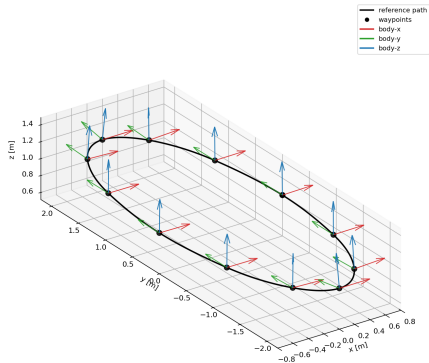


Planned Trajectories — Corkscrew / Screw

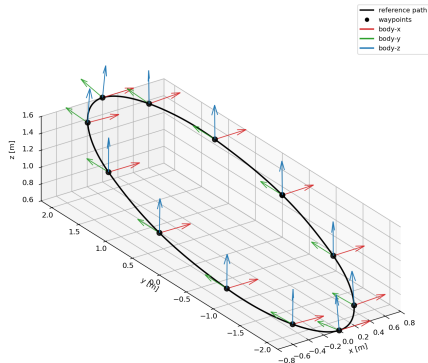


Planned Trajectories — Racing: Oval / Tilted Oval

3D trajectory + orientation triads (waypoints)
model / oval



3D trajectory + orientation triads (waypoints)
model / tilted_oval



INDI — Attitude Control via Incremental Sensor Feedback

Core idea, in one sentence

Don't compute the torque you need from scratch — **measure the torque the motors are already producing right now**, and command only the **small correction** needed to close the remaining gap to the desired angular acceleration.

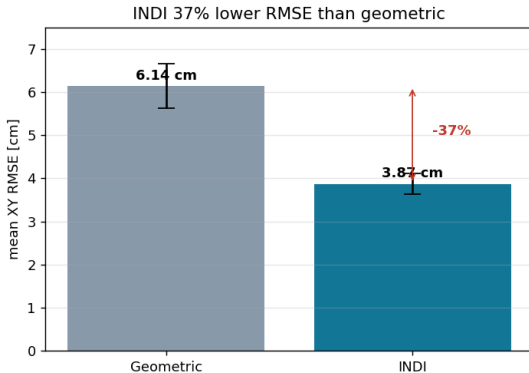
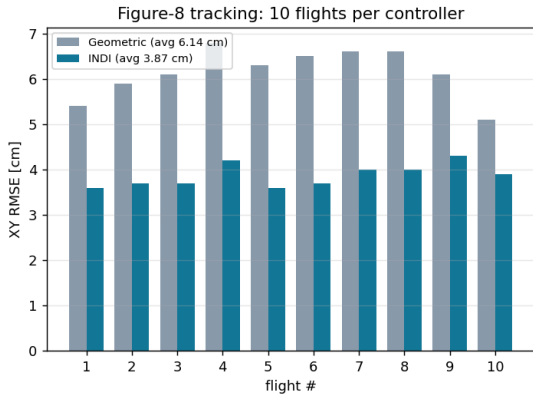
- **Classical (global) nonlinear dynamic inversion** computes the *entire* control input from a complete dynamics model every step — get the model wrong and the whole command is wrong.
- **INDI** (Tal & Karaman 2020) instead measures where the system already is and only asks the model for the small step from there to the target:

$$\underbrace{\tau}_{\text{command}} = \underbrace{\tau_{\text{current}}}_{\text{measured, RPM}^2} + \underbrace{J(\alpha_{\text{ref}} - \alpha_{\text{meas}})}_{\text{small correction, from model}}$$

(α_{meas} : measured from gyro; α_{ref} : desired angular acceleration.)

- Only that *small correction* needs the model (effectiveness J , thrust constants) — sensors answer “where am I now,” the model answers only “how much does a small extra input change things.”
- **Advantage over standard controllers**: unlike PID (reacts after the fact) or pure model-based control (only as accurate as its inertia model), INDI stays accurate even when that model is wrong.

Geometric vs. INDI — Standard CF2.1, 10 Flights Each



37% lower RMSE with INDI.

Figure-8 — Four-Way Comparison, Mean-Closest Flight

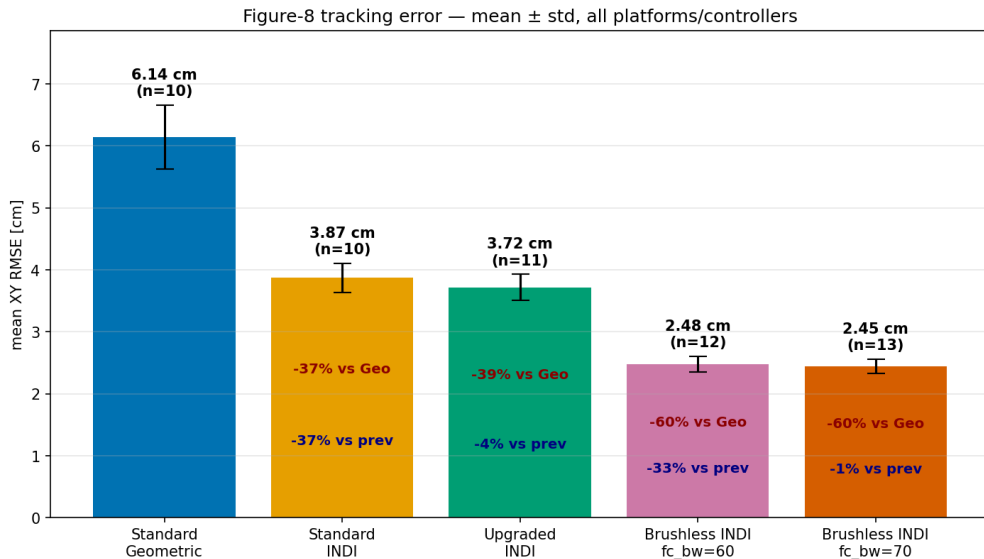


Figure-8 — Best Flight, All Four Modes: Overview

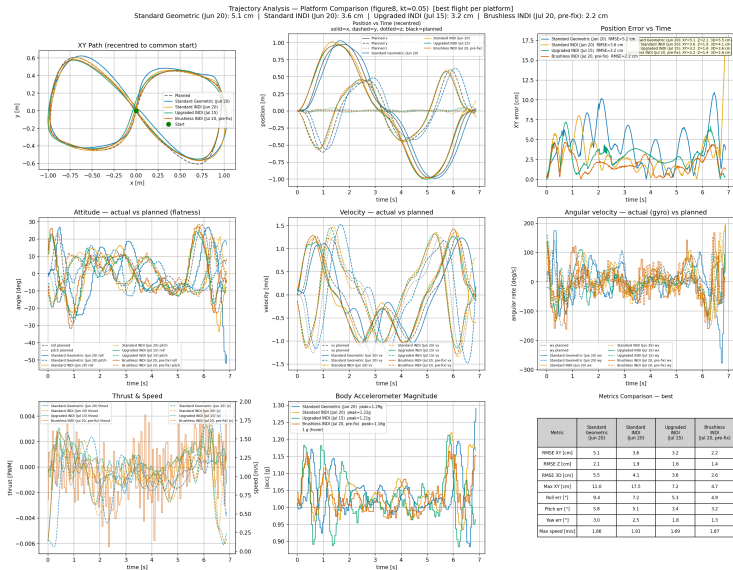


Figure-8 — Best Flight, All Four Modes: Per-Axis

Per-axis Tracking Subplots — Platform Comparison (figure8, kt=0.05) [best]

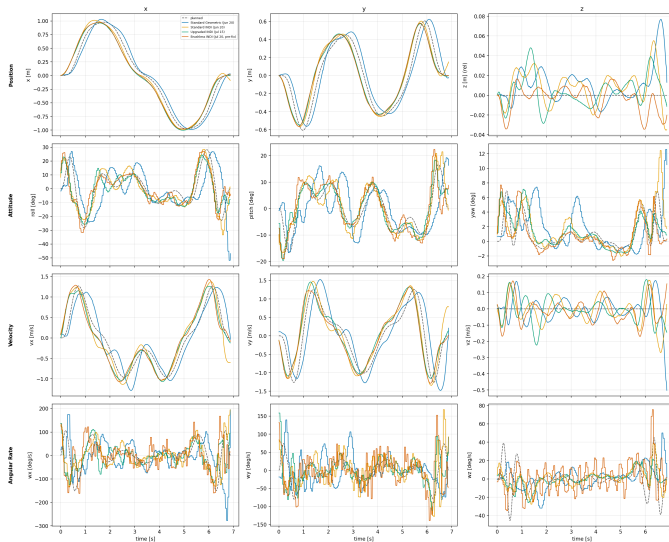
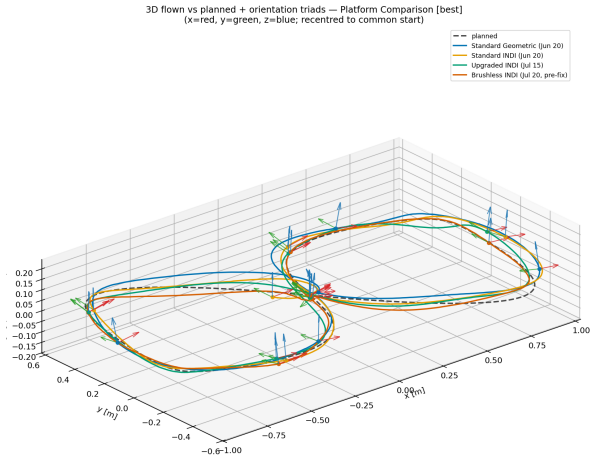


Figure-8 — Best Flight, All Four Modes: 3D Orientation

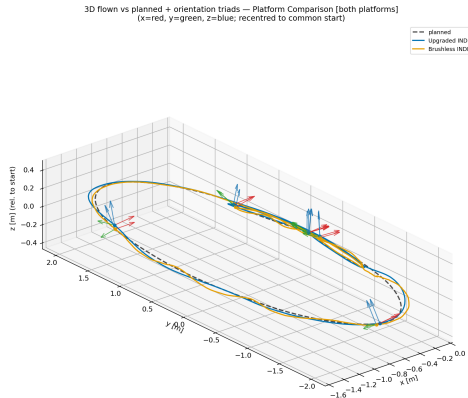


Oval kt Sweep — Both Platforms

kt	XY / Z RMSE		Roll / Pitch / Yaw err		Max speed [m/s]	
	Upgraded	Brushless	Upgraded	Brushless	Upgraded	Brushless
0.05	2.1 / 0.8 cm	2.3 / 0.5 cm	1.7/1.3/0.8°	2.2/2.9/0.9°	1.10	1.22
0.1	2.8 / 0.9 cm	1.3 / 0.7 cm	2.3/1.5/0.7°	3.1/1.6/0.3°	1.33	1.32
0.2	2.9 / 1.3 cm	2.5 / 1.1 cm	3.3/1.5/1.0°	2.6/1.3/0.4°	1.61	1.51
0.3	3.0 / 1.8 cm	2.8 / 1.0 cm	3.2/1.7/1.4°	2.9/1.5/0.5°	1.76	1.72
0.4	3.2 / 2.1 cm	3.2 / 1.4 cm	4.4/2.1/1.9°	10.5/6.1/2.0°	1.97	2.25
0.5	crash (5/5)	3.1 / 1.6 cm	—	5.5/2.7/1.4°	—	2.20
0.6	crash	not flown	—	—	—	—
0.7	not flown	crash	—	—	—	—

Working ceiling: upgraded ~kt 0.3–0.4, brushless ~kt 0.5–0.7.

Oval $kt=0.4$ — Both Platforms: 3D Orientation



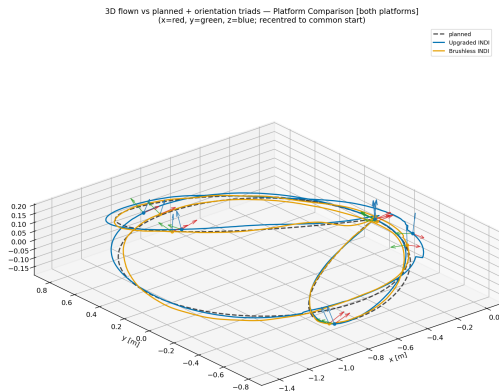
XY/Z RMSE: upgraded 3.2/2.1 cm, brushless 3.2/1.4 cm — comparable XY, brushless tighter on Z.

Circle kt Sweep — Both Platforms

kt	XY / Z RMSE		Roll / Pitch / Yaw err		Max speed [m/s]	
	Upgraded	Brushless	Upgraded	Brushless	Upgraded	Brushless
0.05	1.7 / 0.8 cm	1.9 / 0.6 cm	2.1/1.8/0.8°	1.9/1.4/0.6°	1.03	1.21
0.1	1.9 / 1.3 cm	1.7 / 0.6 cm	1.9/1.8/1.1°	4.2/3.9/0.7°	1.15	1.50
0.2	2.3 / 1.2 cm	6.1 / 1.8 cm	2.2/1.8/1.4°	14.5/13.2/3.6°	1.35	1.71
0.3	2.2 / 1.5 cm	1.4 / 0.8 cm	2.3/2.1/1.9°	3.1/2.7/0.6°	1.53	1.75
0.4	2.7 / 1.8 cm	4.2 / 1.4 cm	3.5/3.6/2.5°	10.0/9.3/2.4°	1.73	1.84
0.5	2.8 / 1.6 cm	3.7 / 1.2 cm	3.5/3.5/2.6°	7.9/7.5/2.0°	1.84	1.88
0.6	3.7 / 2.4 cm	1.9 / 1.0 cm	3.1/3.3/2.7°	4.7/4.9/1.5°	1.98	1.84
0.7	4.2 / 1.7 cm	2.4 / 1.2 cm	4.4/5.6/3.4°	7.3/6.8/2.3°	2.11	2.13
0.8	3.3 / 2.5 cm	crash	4.8/5.3/3.6°	—	2.37	—
0.9	crash	crash	—	—	—	—
1.0	3.0 / 3.6 cm	not flown	4.9/4.0/4.6°	—	2.48	—

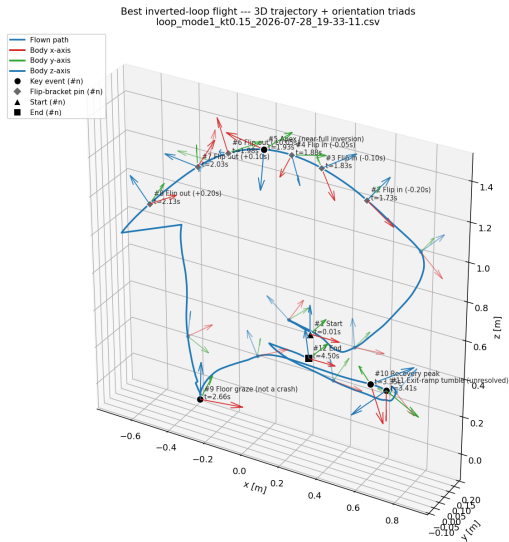
Both platforms show the same pattern: a clean ceiling, then a marginal/inconsistent zone — not a sharp wall, not a platform-specific quirk.

Circle $kt=0.7$ — Both Platforms: 3D Orientation



XY/Z RMSE: upgraded 4.2/1.7 cm, brushless 2.4/1.2 cm — brushless tighter on both axes here.

Best Inverted-Loop Flight — 3D Trajectory



Flight footage.

Summary

- ➊ **Motion planning:** built a pipeline that generates optimal trajectories for the drone — the foundation everything else in this project flies on.
- ➋ **Baseline:** established and tuned a geometric controller on the standard CF2.1 for optimal tracking error, as the reference every other mode is measured against.
- ➌ **INDI:** implemented and deployed full INDI on **three platforms** — standard, thrust-upgraded, and brushless CF21BL.
- ➍ **Comparison:** fair, apples-to-apples evaluation across all four modes — INDI beats geometric on every platform, and brushless beats standard/upgraded on raw tracking (best figure-8 RMSE of all four, 2.45 cm).
- ➎ **Artifacts:** flights across the full trajectory library (figure-8, oval, circle, ...) on multiple platforms, at multiple speeds — the concrete outcome of the project.
- ➏ **Limitations:** an unresolved brushless attitude resonance and the inverted-loop maneuver (flip achieved, recovery not yet made to work) — both well-characterized open problems, not unexplained gaps.